# RhodoSim++

User manual

October 2026

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## 1 Introduction

RhodoSim++ is a simulation and design tool for rhodotron electron accelerators that runs entirely in a web browser. It brings together two programs:

- **RhodoSim**, which tracks electrons through the rhodotron's coaxial cavity and its deflecting magnets. RhodoSim++ is compatible with the C++ RhodoSim by F. Er and G. Unel: it reads and writes the same `config.ini` files, and one of its integrators reproduces the C++ code step for step.
- **The Rhodotron Cavity Profiler**, which designs the RF cavity: it solves the cavity's resonant mode for a chosen cross-section and hands the solved field to the tracking.

The interface is available in English and Turkish. Installation does not need any software packages.

If you use this software, please cite: Er, Muhammet Furkan. "Design and simulation software for rhodotron type electron accelerators." MSc thesis, Boğaziçi University, 2024. We acknowledge support from TÜBİTAK under grant 122F304.

## 2 Getting started

Open `rhodosim.html`[1] in a recent web browser (Chrome, Edge, Firefox or Safari). Everything runs on your computer, without a network connection (online, the page uses its web fonts; offline, the browser's own). Your settings are kept in the browser between visits.

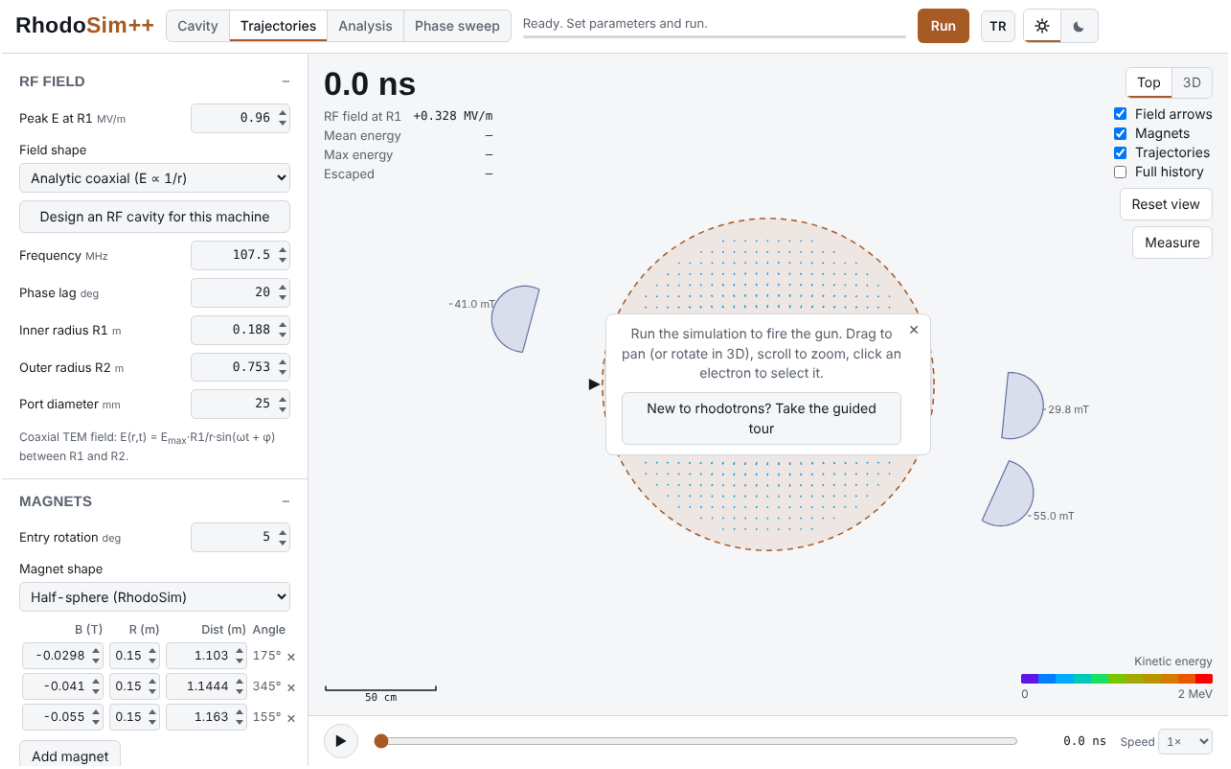



**Figure 1:** RhodoSim++ when first opened: the header with its four tabs, the settings sidebar on the left, the trajectory view, and the welcome box.

The **header** holds the four tabs (Cavity, Trajectories, Analysis, Phase sweep), a status line, the Run simulation button, the language switch (TR/EN) and the Day/Night theme. The **sidebar** holds the machine's settings and is shared by the three beam tabs; the Cavity tab has its own controls. Below the trajectory view, the **timeline** plays a finished run back.

The welcome box can be closed with its ×; it stays closed. If you are new to rhodotrons, start with the Guided tour (header button, or the welcome box): it introduces the machine step by step, runs small

[1] A recent version can be obtained at `https://erkcan.github.io/RhodoSimWeb/Web/rhodosim.html`.

experiments, and restores your own settings when you leave it. The Cavity tab has two tours of its own (Section 6).

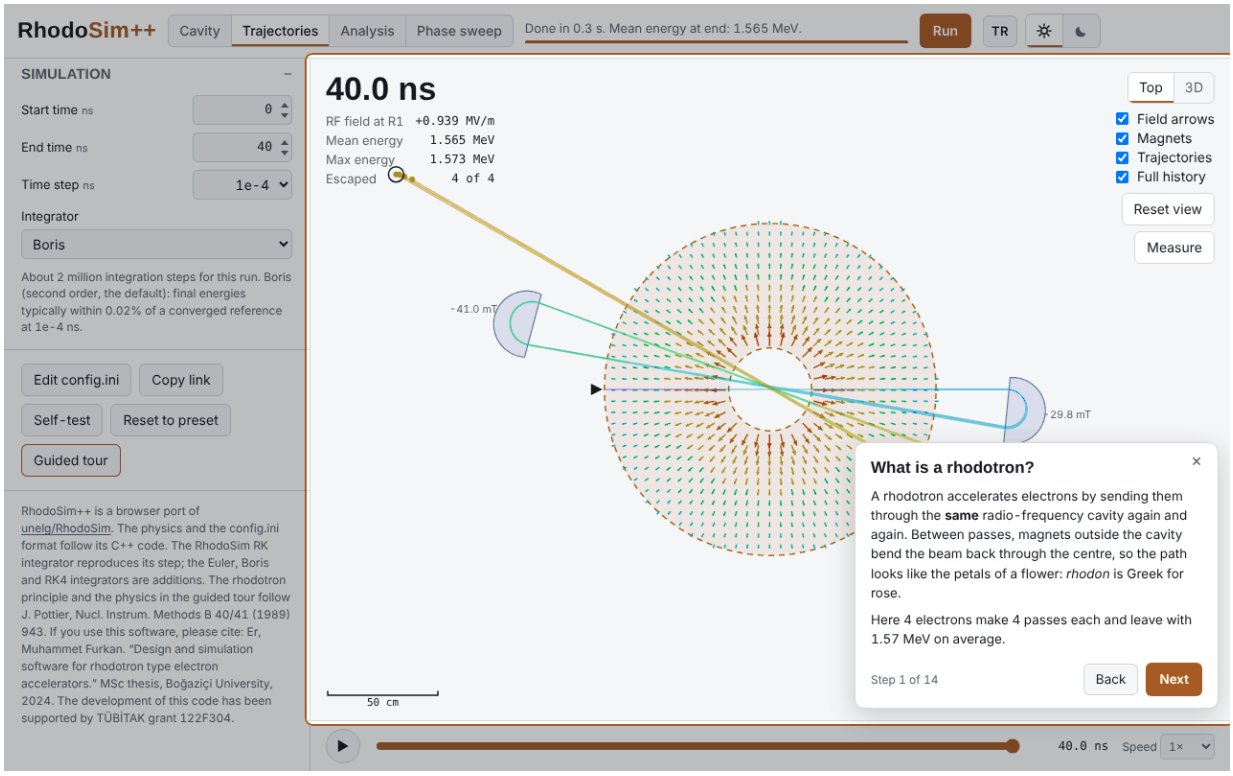



**Figure 2:** The guided tour. Each step explains one aspect of the rhodotron and shows it in the simulation; some steps invite you to try a change yourself.

## 3 Trajectories

The Trajectories tab simulates the beam: electrons from the gun cross the cavity, gain energy from its RF field, leave it, are turned around by a magnet, and cross again, pass after pass. Press Run simulation; the status line reports the run's duration and the mean energy at the end.

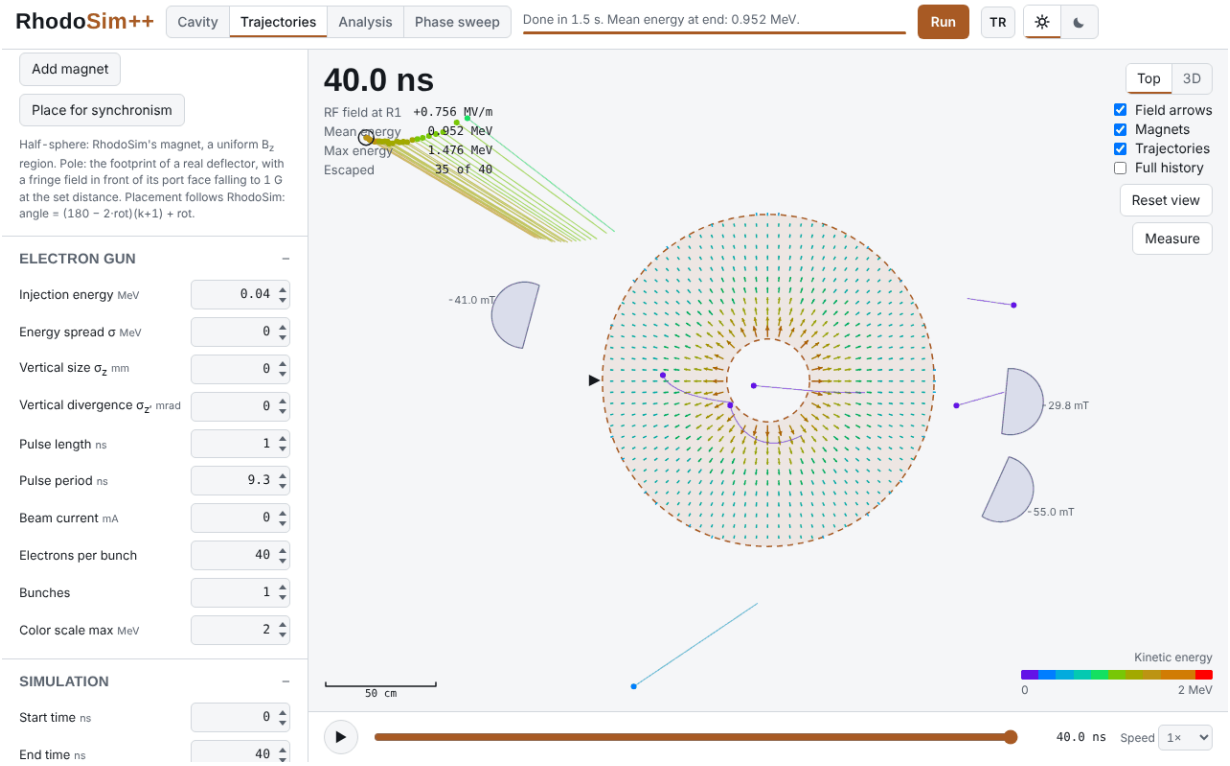



**Figure 3:** A finished run in the top view. The trajectories are coloured by kinetic energy (the scale is at the bottom right); the readouts at the top left belong to the moment shown on the timeline.

## 3.1 Settings

**RF field.** Peak E at R1 (MV/m) is the field amplitude at the inner conductor; Frequency (MHz), Phase lag (degrees), and the conductor radii Inner radius R1 and Outer radius R2 (m) define the cavity. Port diameter (mm) is the opening in the outer conductor through which the beam passes (25 mm for KF-25 ports). Field shape is either Analytic coaxial ($E \propto 1/r$), the field of an ideal coaxial line, or Solved cavity (Cavity tab), the field of a cavity designed in the Cavity tab (Section 7).

**Magnets.** Each row of the table is one magnet: its field B (T), radius R (m) and distance Dist (m) from the cavity's centre; its angle follows from the Entry rotation. Add magnet appends one. Magnet shape is Half-sphere (RhodoSim), C++ RhodoSim's model (a uniform field in a half-ball), or Pole, fringe field, the footprint of a real deflector with a fringe field in front of its face, whose extent is set by Field 1 G at (m); pole magnets also have a Pole gap (m). Place for synchronism moves each magnet along its line from the centre until the reference electron crosses the centre at the same RF phase on every pass; Undo restores the earlier distances.

**Electron gun.** Injection energy and Energy spread $\sigma$ (MeV), Vertical size $\sigma_z$ (mm) and Vertical divergence $\sigma_{z'}$ (mrad), Pulse length and Pulse period (ns), Beam current (mA), Electrons per bunch and Bunches. The beam current does not change the tracking: it sets the space-charge guide in the Analysis tab (Section 4). With both vertical spreads at zero every electron stays in the mid-plane; any spread makes the simulation three-dimensional (Section 3.4).

**Simulation.** Start time and End time (ns), the Time step, and the Integrator: Euler (first order), Boris (the default, second order), RK4 (fourth order), or RhodoSim RK, C++ RhodoSim's own scheme, which reproduces the C++ code step for step (Section 8). A note under the integrator estimates the run's cost. Edit

config.ini shows the settings as a RhodoSim configuration file, Self-test checks the engine (Section 11), and Reset to preset restores the example machine.

## 3.2 The view

Top and 3D switch the view. Drag to pan (or rotate in 3D), scroll to zoom, and click an electron to select it; its passes appear in the Analysis tab. The check boxes show or hide Field arrows, Magnets, Trajectories and the Full history; with a solved cavity field, Cavity wireframe draws the cavity's walls. Color scale max (MeV) sets the top of the energy colour scale.

The readouts show the RF field at R1, the Mean energy and Max energy, and how many electrons have Escaped, were Stopped (a numerical breakdown) or were Lost on the machine's apertures. Two labels may appear over the view: Non-physical setup, when a run was made despite a warning (Section 7), and Trajectories from earlier settings: Run to update, when the settings have changed since the run shown. The view always draws the current settings; the trajectories stay those of the last run until you run again.

## 3.3 Measuring

In the top view, the pointer's position is shown at the lower left, in metres. Measure (or the M key) turns a drag into a ruler; its readout gives the distance, $\Delta x$, $\Delta y$ and the angle against the $x$ axis. The ruler is anchored to the machine, so it follows zooming and panning, and it stays until the next measurement, Escape, or leaving measure mode. Holding Shift snaps its ends to the cavity's centre, the nearest point on either conductor, the magnets' centres and the ends of their faces, and the electrons at the moment shown. Measuring is offered only in the top view.

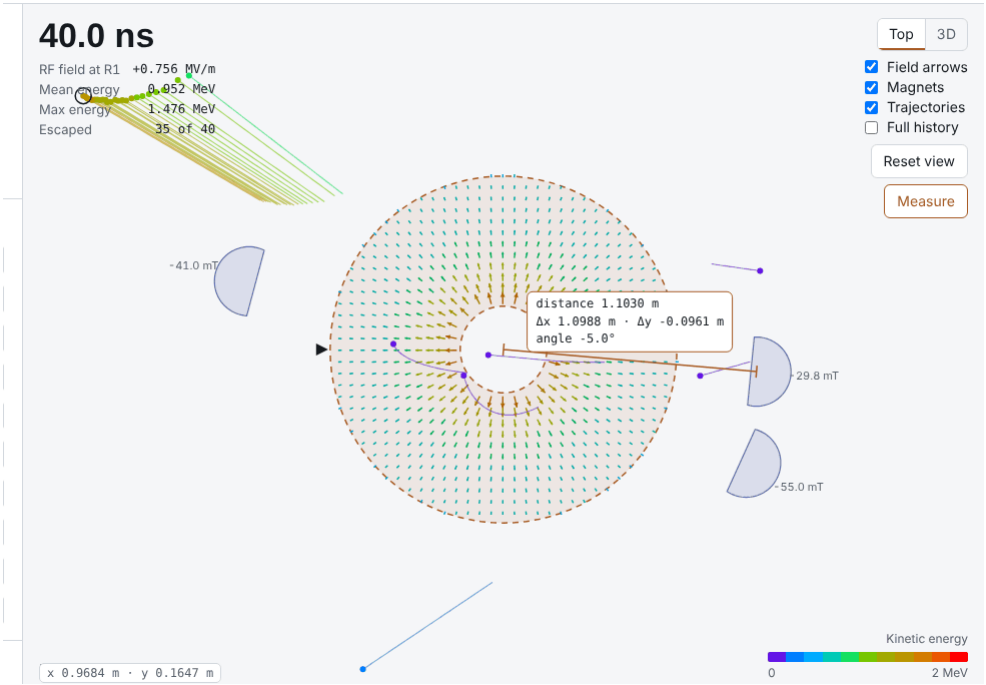



**Figure 4:** A ruler from the cavity's centre to the first magnet's centre, both ends snapped with Shift.

## 3.4 Three-dimensional tracking

With a vertical spread at the gun, electrons leave the mid-plane. The RF field and its magnetic field act on them there, pole magnets focus them vertically at their faces, and electrons that rise too far are lost: at a port, when they cross the outer conductor above or below half the Port diameter, or on a pole, above

or below half the Pole gap. In the 3D view, the ×1, ×10 and ×50 buttons magnify the beam's heights (millimetres in a metre-sized machine); the cavity stays at true scale. With pole magnets, the 3D view also draws each magnet's poles above and below the gap, and the ports' aperture as rings on the outer conductor; their heights follow the same magnification.

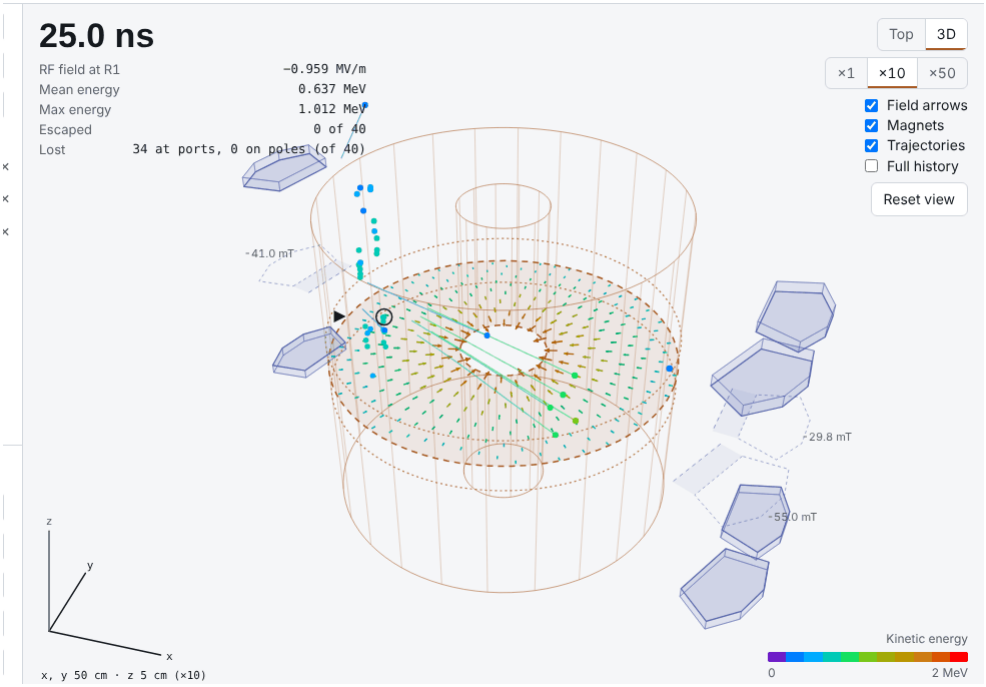



**Figure 5:** A run with a vertical spread and pole magnets in the 3D view, heights magnified ten times: the poles above and below the gap, the port rings on the outer conductor, and lost electrons where they hit.

# 4 Analysis

The Analysis tab studies the last run. Energy over time plots the selected electron's energy, or the bunch's; Energy distribution shows the energies at a chosen moment as a histogram with its statistics. Passes through the cavity lists the selected electron's passes, and All electrons, pass by pass summarises the bunch: for each pass, how many electrons completed it, their energy after it, the RF phase at the centre, and the loop's length in RF periods.

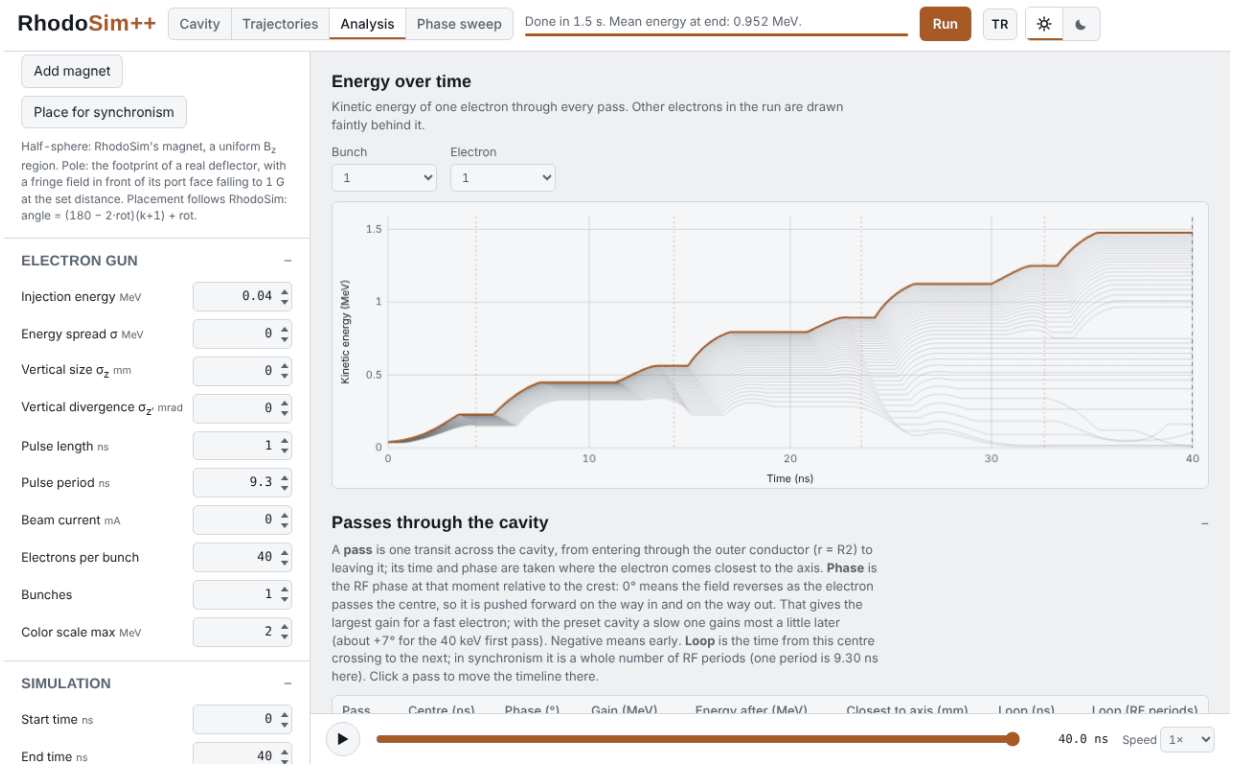



**Figure 6:** The Analysis tab: the energy over time and the selected electron's passes through the cavity; the energy distribution and the bunch's summary follow further down.

| Pass | Electrons | Energy after (MeV) | Phase (°) | Loop (RF periods) |
|---|---|---|---|---|
| 1 | 40 of 40 | 0.4112 ± 0.0357 | 28.1 ± 12.2 | 1.077 |
| 2 | 40 of 40 | 0.6259 ± 0.1456 | 55.8 ± 18.9 | 1.025 |
| 3 | 40 of 40 | 0.7765 ± 0.3258 | 63.5 ± 29.5 | 1.022 |
| 4 | 25 of 40 | 1.2849 ± 0.1498 | 37.9 ± 10.8 | – |

**Figure 7:** The bunch, pass by pass. A run with vertical motion adds the vertical beam size and divergence at each pass's centre.

When the run has vertical motion, the summary adds the vertical size ($\sigma$ height, mm) and divergence ($\sigma$ angle, mrad) at each pass's centre, and Vertical phase space plots every electron's height against its angle for a chosen pass, with the losses by cause.

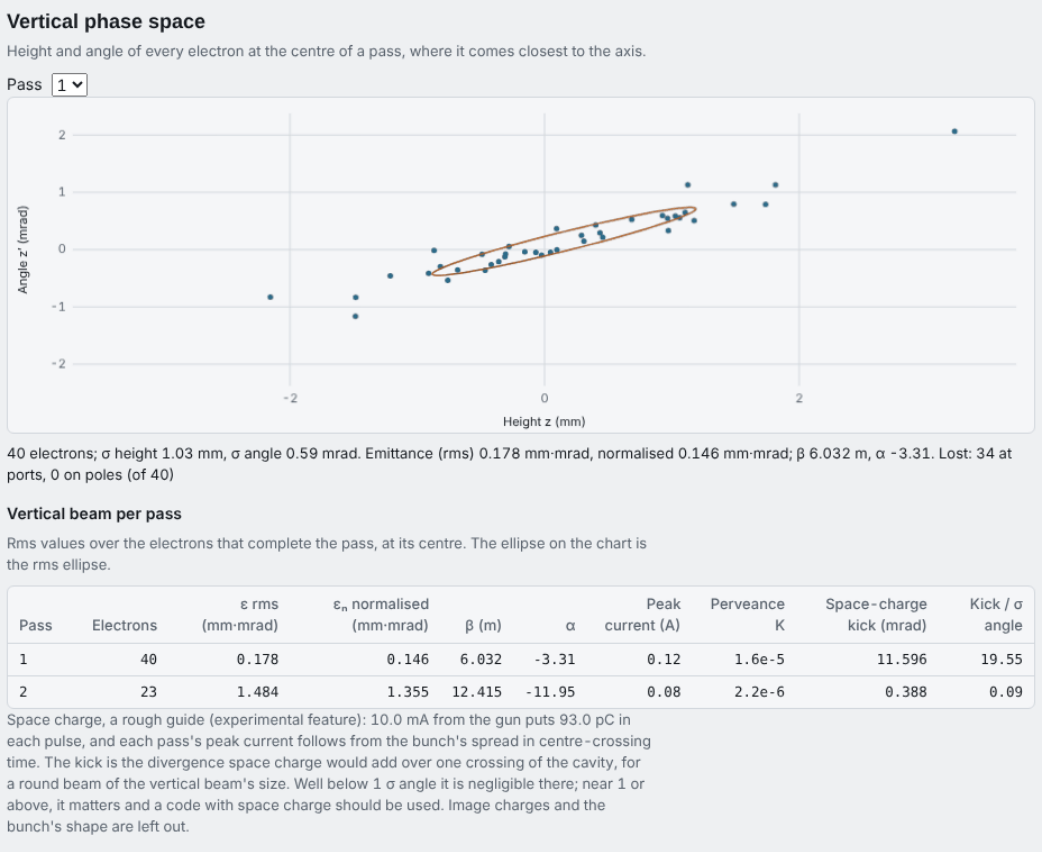



**Figure 8:** The vertical phase space at the first pass, with the rms ellipse, and the vertical beam per pass with the space-charge guide for 10 mA. The upward tilt is a diverging beam: higher electrons are moving upwards.

Below the chart, Vertical beam per pass gives the vertical beam's figures of merit at each pass's centre, over the electrons that complete the pass, read at the centre itself (between the logged frames). With $z$ the height and $z'$ the angle, and $\langle\cdot\rangle$ the mean over the electrons after removing the beam's centre,

$$\varepsilon = \sqrt{\langle z^2\rangle\langle z'^2\rangle - \langle zz'\rangle^2}, \qquad \beta = \frac{\langle z^2\rangle}{\varepsilon}, \qquad \alpha = -\frac{\langle zz'\rangle}{\varepsilon}, \qquad \gamma = \frac{\langle z'^2\rangle}{\varepsilon}.$$

$\varepsilon$ is the rms emittance (mm mrad, the area of the rms ellipse over $\pi$); $\beta$, $\alpha$ and $\gamma$ are the Twiss parameters of that ellipse. The emittance shrinks as the beam gains energy; the normalised emittance $\varepsilon_n$ takes this out by using each electron's own momentum, $u = \beta\gamma z'$ in place of $z'$, so an energy spread is not counted as emittance. Without collective effects $\varepsilon_n$ stays constant through the machine, apart from losses and the non-linear fields, so a growing $\varepsilon_n$ shows where the beam's quality is lost. The chart draws the rms ellipse of the pass over the electrons. Only the vertical plane is measured: in the mid-plane the bunch's spread is set by the energy and the RF phase, which the pass-by-pass summary already gives.

**Space charge, a rough guide (experimental feature).** The tracking leaves out the electrons' fields on each other. To judge whether they matter, set Beam current (mA, the gun's average current; 0 turns the guide off). Each gun pulse then carries $Q = I\,T_{\text{gun}}$, with $T_{\text{gun}}$ the Pulse period, and the table adds for each pass:

- the peak current $I_{\text{pk}} = Q/(\sqrt{2\pi}\,\sigma_t)$, with $\sigma_t$ the rms spread of the electrons' centre-crossing times within a bunch (a Gaussian bunch);
- the generalised perveance $K = 2I_{\text{pk}}/(I_0(\beta\gamma)^3)$, $I_0 = 17.045\,\text{kA}$;
- the space-charge kick $KL/a$ (mrad): the divergence the beam's own field would add over one crossing of the cavity, $L = 2R_2$, for a round beam of radius $a = 2\sigma_z$;

- the ratio of this kick to the beam's rms angle.

Well below 1, space charge is negligible at that pass; near 1 or above it matters, and a code that includes space charge should be used. Image charges in the walls, the bunch's actual shape and the coupling between the planes are left out: the guide says whether to worry, not by how much. The low-energy first passes are where it bites, because $K$ falls as $(\beta\gamma)^{-3}$.

## 5 Phase sweep

The Phase sweep tab runs the machine over a range of RF phase lags and plots a chosen quantity at a chosen time: Mean energy, Energy spread $\sigma_E$, Position spread $\sigma_R$, or Transmission, the share of electrons not lost. The chart marks the phase with the highest mean energy, the one with the smallest energy spread above 80 % of that, and the current phase lag (dashed). Click a point to adopt its phase lag.

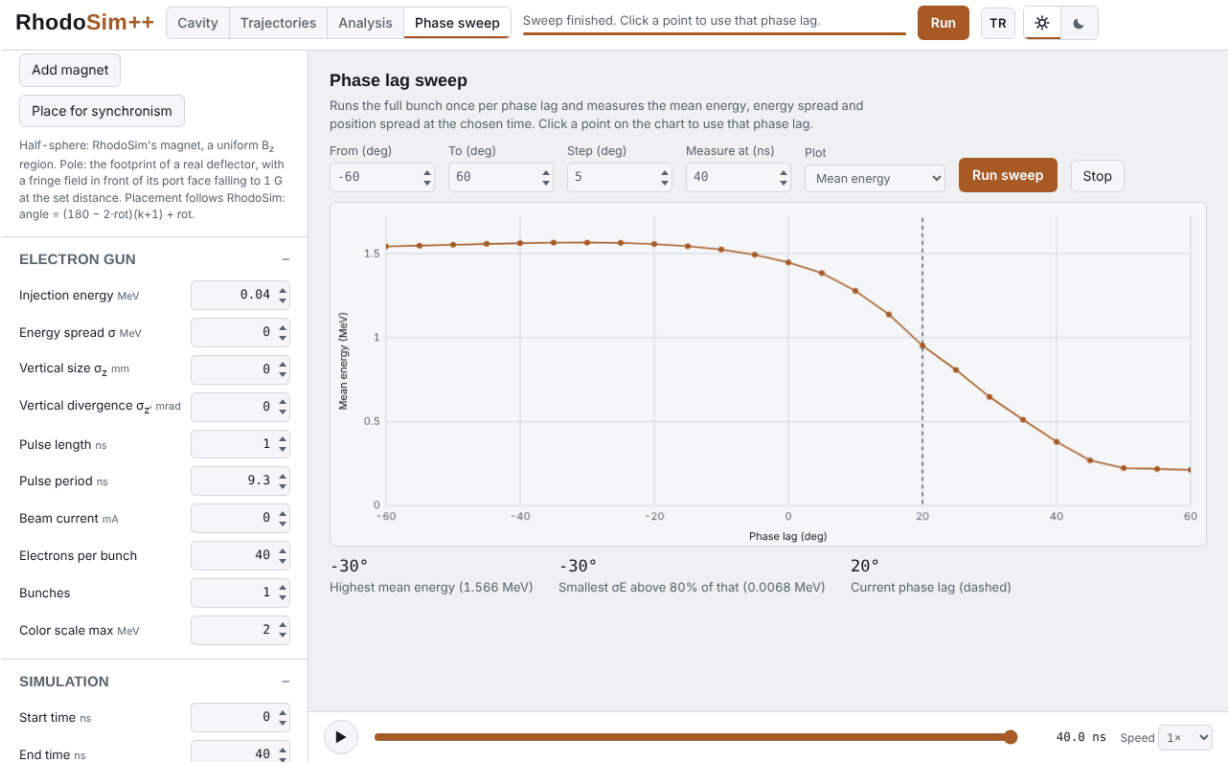



**Figure 9:** A phase sweep from $-60°$ to $60°$ in $5°$ steps.

## 6 The Cavity tab

The Cavity tab designs the RF cavity. A rhodotron cavity is a coaxial line shorted at both ends, a body of revolution; the tab solves its resonant mode in the $r$–$z$ half-section and shows the fields, the cavity's figures of merit, and their accuracy.

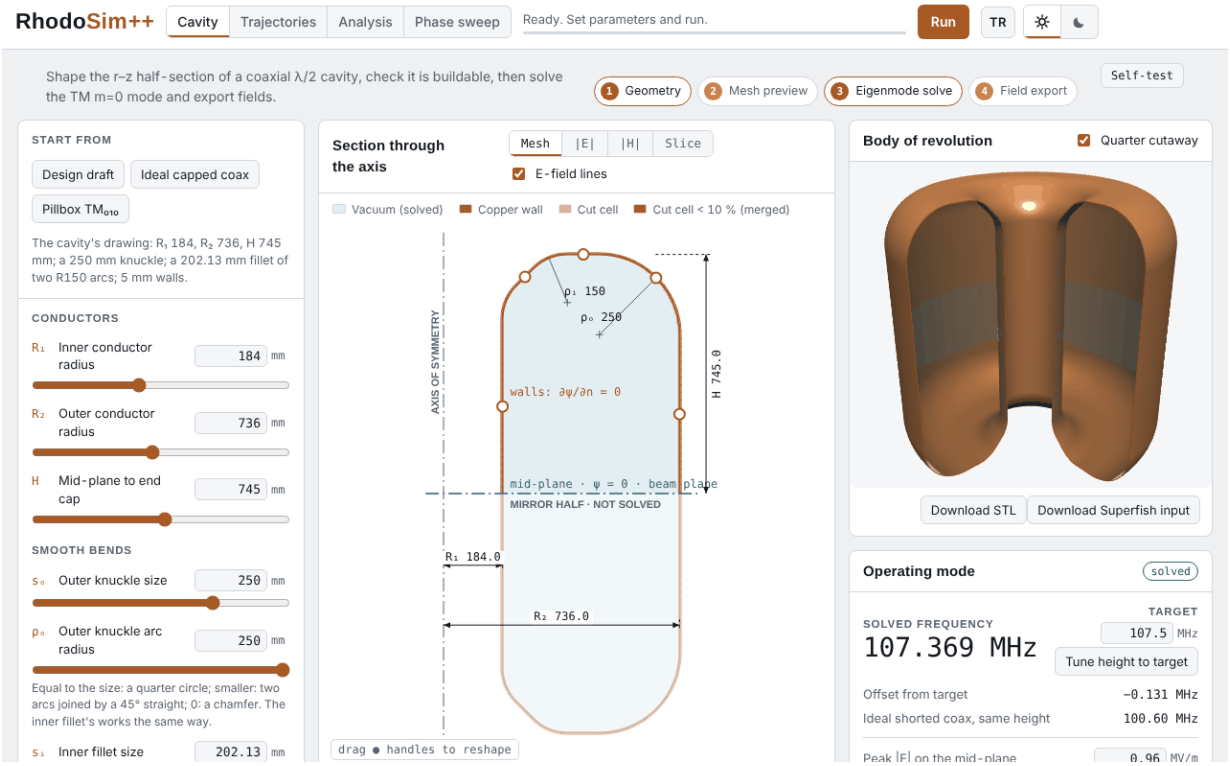



**Figure 10:** The Cavity tab: the parameters, the section through the axis, the 3D view, and the readouts.

**Parameters.** Start from offers example cavities. The section is set by the inner and outer radii $R_1$ and $R_2$, the half-height $H$, the outer corner (the knuckle) and the inner one (the fillet), the wall thickness, and the mesh's cell size $h$. Each corner has a size, how far it reaches along both walls, and an arc radius: equal to the size it is a quarter circle; smaller, two arcs of that radius joined by a 45-degree straight (a rounded chamfer, as on many drawings); 0, a plain chamfer. The arc radius follows the size while the two are equal. The Mid-plane boundary condition is a magnetic wall (the rhodotron's mode) or an electric wall. The mode is solved automatically after a change (Solve eigenmode); the status shows solved, solving, or out of date. A target frequency and Tune height to target adjust $H$ until the cavity resonates at the target.

**The section.** Mesh, |E| and |H| show the mesh or the field magnitudes on the section; Slice shows a horizontal cut through the cavity.

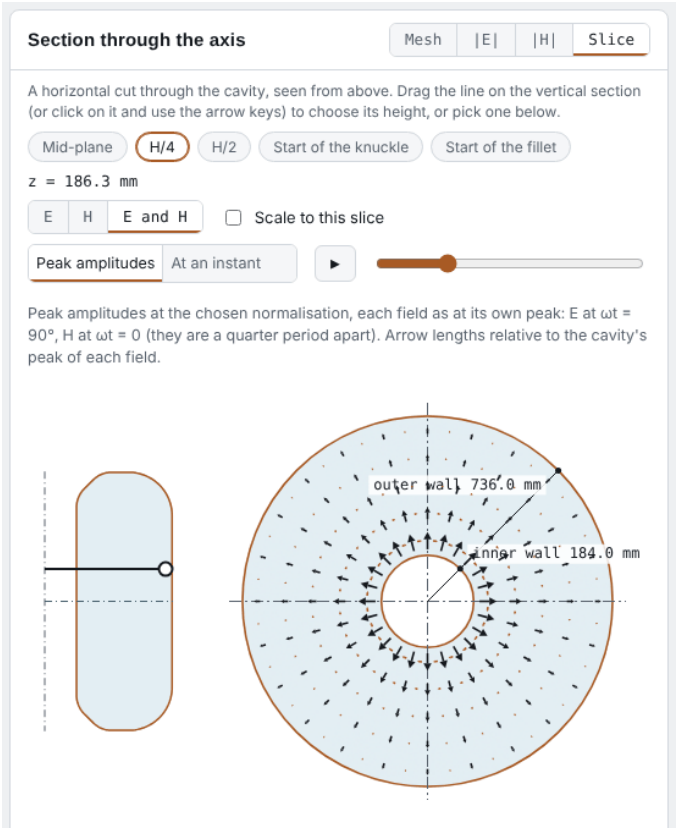



**Figure 11:** The slice view at a quarter of the height. Radial arrows are $E_r$, tangential ones $H_\phi$, and the wash is $E_z$.

In the slice view, drag the height line on the small section (or use the arrow keys), or pick a height: Mid-plane, H/4, H/2, Start of the knuckle, Start of the fillet. E, H or E and H choose the arrows; Scale to this slice rescales them to the slice's own peak. Peak amplitudes shows each field at its own peak (they are a quarter period apart); At an instant shows both at a phase $\omega t$ set by the slider, and the play button animates one RF cycle.

**Readouts.** The frequency, the copper $Q_0$, the gap voltage, the stored energy, the wall loss and the peak surface field, with the mesh's size; Mid-plane field plots $E_r$ along the mid-plane. The 3D view shows the cavity's copper body (a quarter is cut away; untick Quarter cutaway for the whole body): drag to turn it, scroll or pinch to zoom, right-drag (Shift-drag, or two fingers) to move it. Download STL saves the full body as an STL file in millimetres. Download Superfish input writes the cavity as an Automesh (`.af`) input file for LANL's Poisson Superfish: the half cavity above the mid-plane in centimetres, its boundary conditions, the solved frequency and, with a solution, a normalisation that makes Superfish's fields match RhodoSim++'s.

**Mesh convergence.** Run convergence study solves the cavity at six cell sizes and extrapolates each quantity to an infinitely fine mesh. The table gives each quantity at your mesh, its extrapolated value, the estimated error at your mesh, and the observed order of convergence.

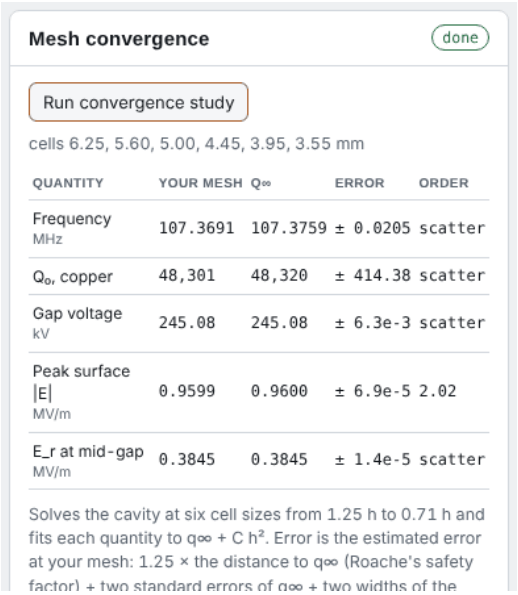



**Figure 12:** A convergence study. The error column is the uncertainty to quote with your mesh's values.

**Field export.** Export fields writes the solved field on a Cartesian grid for RhodoSim, as text or binary, in CST's nine-column format, or the native $r$–$z$ grid; Copy config.ini lines gives the matching RhodoSim settings.

**Guided tours.** Two tours of their own sit at the bottom of the parameters panel, both on the design draft at its operating field, with every number read from the solver (a derived one shows its arithmetic on hover). Inside the cavity, for newcomers, shows without changing anything: the coaxial cavity and its shape, the resonance against the quarter wave $c/4H$, where the electric and magnetic fields are, the field the beam sees in the mid-plane, $E$ and $H$ a quarter period apart, the wall loss in kilowatts and $Q_0$, how far to trust the mesh, and the hand-off to Trajectories. Designing a cavity, for students, changes the design step by step: the solver checked on the ideal coax, the height setting the frequency, what rounding the corners does, and the choice of $R_1$ (voltage against power). Both tours restore your own design when they end and store nothing of theirs; each resumes at its own step.

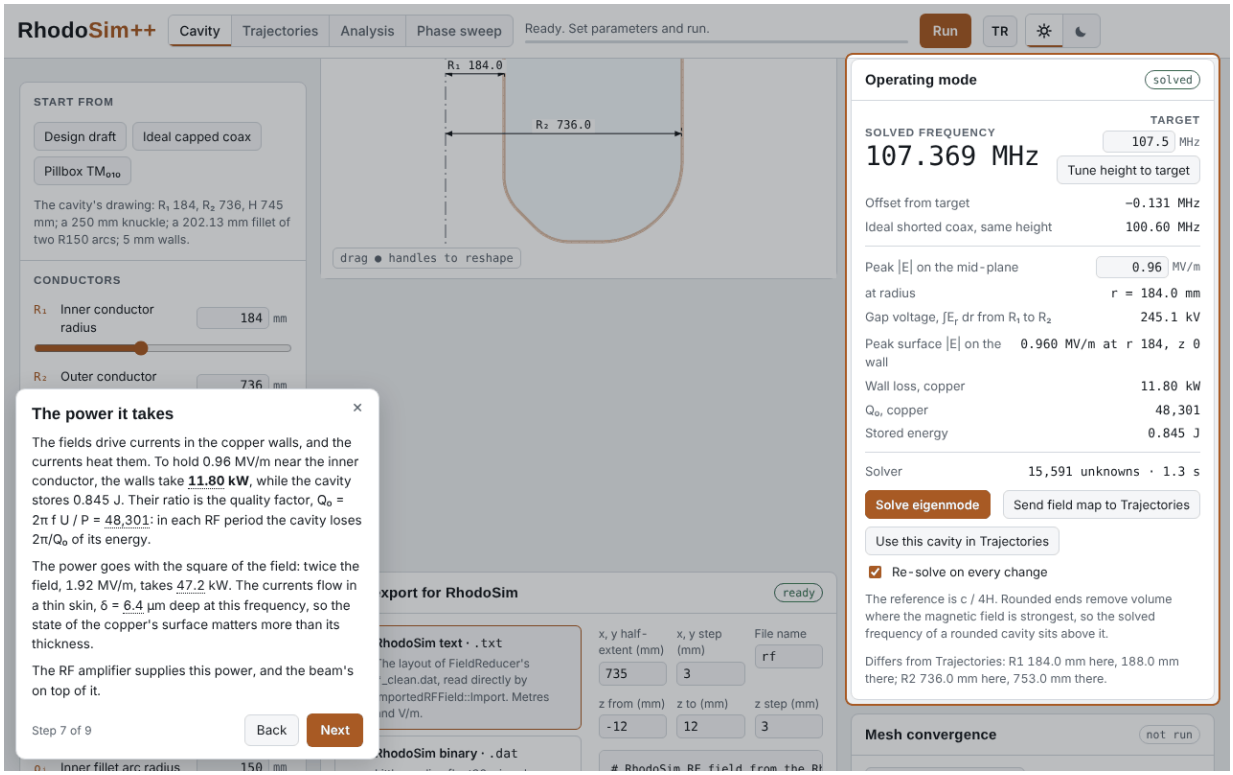



**Figure 13:** Inside the cavity: the power step, with the wall loss and $Q_0$ for the design draft at 0.96 MV/m.

## 7 Designing a machine

RhodoSim++ has two uses. For learning and simulation, start in Trajectories and explore. For design, the usual path is frequency, cavity, magnets, beam, iterating. The tabs keep their own values and never change each other on their own; values move only when you ask:

- Design an RF cavity for this machine (Trajectories) opens the Cavity tab with the machine's radii and RF frequency as the target.
- Use this cavity in Trajectories (Cavity) sends the radii, the resonant frequency and the solved field to the beam, and selects the solved field. Send field map to Trajectories sends the field alone.

When the tabs disagree, warnings say so. Amber warnings point out differences: the tabs' radii or frequencies differ, the cavity does not resonate at the RF frequency, or the magnets were placed for another $R_2$ (Keep them accepts them). Red warnings mark a setup that is not physical: a field map for another cavity, a magnet inside the outer conductor, or two magnets whose fields overlap. A non-physical setup runs only after confirmation, and its results are labelled.

## 8 The physics model

**Motion.** Electrons obey the relativistic Lorentz force,

$$\frac{d\mathbf{p}}{dt} = q\,(\mathbf{E} + \mathbf{v} \times \mathbf{B}), \tag{1}$$

in RhodoSim's units (ns, m, MV/m, T, MeV). An electron starts moving at the first time step after it is fired, as in C++ RhodoSim.

**RF field.** The analytic field is the $\lambda/2$ coaxial mode,

$$E_r = E_{\max}\,\frac{R_1}{r}\cos(kz)\,\sin(\omega t+\varphi), \qquad B_\phi = -\frac{E_{\max}}{c}\,\frac{R_1}{r}\,\sin(kz)\,\cos(\omega t+\varphi), \qquad k=\frac{\omega}{c}, \tag{2}$$

between the conductors; in the mid-plane ($z=0$) only $E_r$ remains. The solved field replaces it with the Cavity tab's map of $E_r$, $E_z$ and $B_\phi$ over the whole cavity.

**Magnets.** A half-sphere magnet has a uniform $B_z$ inside a ball of radius $R$, beyond a face plane $R/5$ in front of its centre. A pole magnet follows a deflector drawing; in front of its face the field rises as $B/(1+e^{-d/\lambda})$, with $\lambda$ set so that $|B| = 1\,\mathrm{G}$ at the chosen distance. Off the mid-plane the fringe carries the components Maxwell's equations require, $B_d = z\,\partial B_z/\partial d$ to first order, which focus the beam vertically at the face; the edge focusing follows Brown's formula of accelerator optics.

**Integrators.** Four update steps are offered; $h$ is the time step and $\mathbf{u}=\gamma\mathbf{v}$ the momentum per unit mass.

- **Euler** (first order): the force at the start of the step changes the momentum, and the new velocity moves the electron: $\mathbf{u}_{n+1} = \mathbf{u}_n + h\,\frac{q}{m}\big(\mathbf{E}+\mathbf{v}_n\times\mathbf{B}\big)(\mathbf{x}_n,t_n)$, $\mathbf{x}_{n+1}=\mathbf{x}_n+h\,\mathbf{v}_{n+1}$. One field evaluation per step; the error falls in proportion to the step. It cannot exceed the speed of light, but in a magnet its rotation is not exact and it gains energy slowly. It is there to show what a first-order integrator does.
- **Boris** (the default, second order): half a step of drift, a kick with the field at the middle of the step (half the electric kick, a rotation of exact length in the magnetic field, the other half of the electric kick), and the other half of the drift. In a magnetic field alone it keeps the energy exactly.
- **RK4** (fourth order): the classical Runge–Kutta scheme on $(\mathbf{x},\mathbf{u})$, four field evaluations per step.
- **RhodoSim RK**: C++ RhodoSim's own step, reproduced step for step, for results that must agree with it. Each step is a drift–kick–drift: the electron moves half a step at its velocity, its velocity changes by an acceleration averaged in the manner of RK4 over four trial states (with the field's time held at the start of the step), and it moves the other half at the new velocity. Second order in a uniform field.

**The cavity solver.** The Cavity tab solves the TM mode ($m=0$) for $\psi = rH_\phi$,

$$\frac{\partial}{\partial r}\left(\frac{1}{r}\frac{\partial\psi}{\partial r}\right)+\frac{\partial}{\partial z}\left(\frac{1}{r}\frac{\partial\psi}{\partial z}\right)+k^2\,\frac{\psi}{r}=0, \tag{3}$$

by finite volumes with cut cells at curved walls (cells less than 10 % open are merged). The fields follow from $\psi$: $H_\phi=\psi/r$, $E_r=-\frac{1}{\omega\varepsilon_0 r}\,\partial\psi/\partial z$, $E_z=\frac{1}{\omega\varepsilon_0 r}\,\partial\psi/\partial r$. At a 5 mm mesh the frequency agrees with exact solutions to a few parts per million.

## 9 Integrator accuracy and cost

The app's four integrators were compared on the preset machine,[2] to give a basis for choosing one: the preset's three half-sphere magnets, 20 electrons fired over 1 ns at 0.04 MeV and tracked for 40 ns, at RF phase lags of $-30^\circ$ and $+20^\circ$, and seven time steps from $10^{-3}$ to $10^{-5}$ ns.

[2] As a link that sets every value (Section 10, *Links*): `https://erkcan.github.io/RhodoSimWeb/Web/rhodosim.html#tab=trajectories&emax=0.96&freq=107.5&phaselag=20&r1=0.188&r2=0.753&magrotation=5&addmagnet=-0.0298,0.15,1.103&addmagnet=-0.041,0.15,1.1444&addmagnet=-0.055,0.15,1.163&ein=0.04&einstd=0&targeten=2&guntime=1&gunperiod=9.3&enum=50&bunchnum=1&starttime=0&endtime=40&dt=0.0001&integrator=boris&magmodel=sphere&fringe1g=0.1&fieldmodel=coax&zsig=0&zpsig=0&portdia=25&polegap=0.1`. The study changed only the number of electrons, the phase lag and the time step.

**Method.** Each result is compared with a reference: a classical fourth-order Runge–Kutta integrator written independently of the program, at $5 \times 10^{-6}$ ns and checked at $10^{-5}$ ns. The program follows C++ RhodoSim's time bookkeeping: time accumulates step by step, an electron starts moving at the first step after it is fired, and the result is the last recorded frame. The reference integrates over exactly that interval for each electron and step, so the comparison measures the integration error alone. A few electrons pass so close to a magnet's edge that the edge decides their fate; their reference does not converge, so they are left out of the errors and counted in the table. The cost is the computing time per electron in Node.js on one core, the best of three runs.

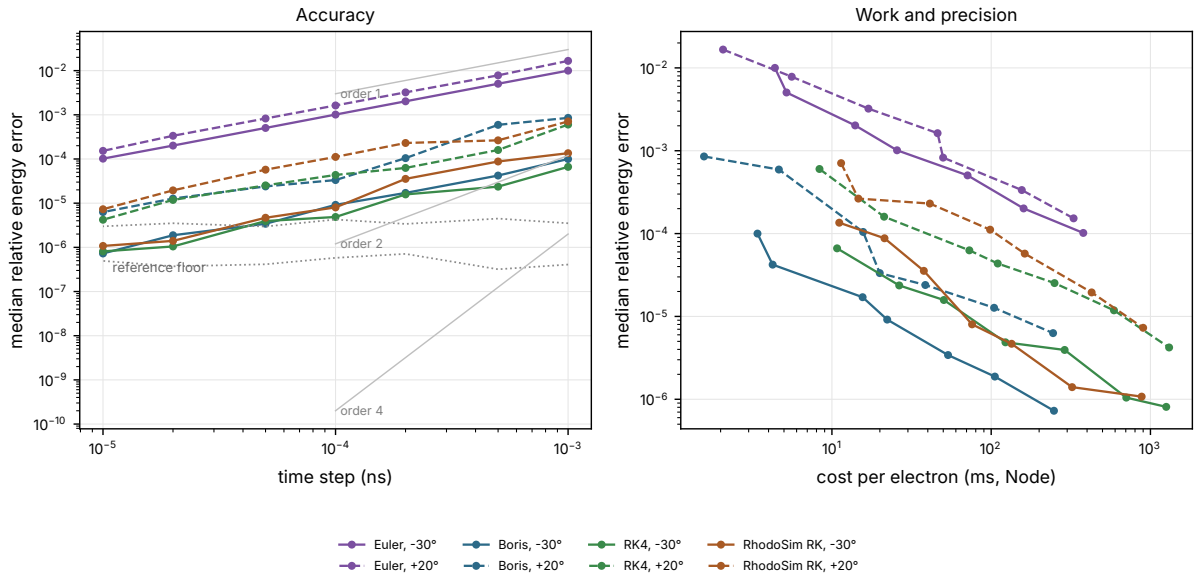



**Figure 14:** Left: the median relative energy error against the time step, for both phase lags (solid $-30°$, dashed $+20°$), with the reference's own floor and guides for orders 1, 2 and 4. Right: the error against the cost.

| Integrator | $\Delta t$ (ns) | median $\lvert\delta E\rvert/E$ $-30°$ | median $\lvert\delta E\rvert/E$ $+20°$ | bias $\langle\delta E\rangle/E$ $-30°$ | median $\lvert\delta \mathbf{r}\rvert$ (mm) $-30°$ | cost (ms/e) |
|---|---|---|---|---|---|---|
| Euler | $10^{-3}$ | 1.0e-02 | 1.7e-02 | +1.0e-02 | 14.13 | 3.2 |
| | $10^{-4}$ | 1.0e-03 | 1.6e-03 | +1.0e-03 | 1.53 | 35.9 |
| | $10^{-5}$ | 1.0e-04 | 1.5e-04 | +1.0e-04 | 0.14 | 353.9 |
| Boris | $10^{-3}$ | 1.0e-04 | 8.5e-04 | -3.5e-05 | 5.85 | 2.5 |
| | $10^{-4}$ | 9.2e-06 | 3.3e-05 | -2.8e-06 | 0.37 | 21.1 |
| | $10^{-5}$ | 7.3e-07 | 6.3e-06 | +6.0e-08 | 0.07 | 246.2 |
| RK4 | $10^{-3}$ | 6.6e-05 | 6.0e-04 | -1.6e-05 | 4.70 | 9.6 |
| | $10^{-4}$ | 4.9e-06 | 4.4e-05 | -8.0e-07 | 0.49 | 116.4 |
| | $10^{-5}$ | 8.1e-07 | 4.2e-06 | +3.9e-07 | 0.05 | 1283.1 |
| RhodoSim RK | $10^{-3}$ | 1.4e-04 | 7.1e-04 | -1.7e-04 | 5.37 | 11.3 |
| | $10^{-4}$ | 8.0e-06 | 1.1e-04 | -5.9e-06 | 0.41 | 87.2 |
| | $10^{-5}$ | 1.1e-06 | 7.3e-06 | -1.1e-06 | 0.05 | 890.3 |

**Table 1:** The four integrators of RhodoSim++'s Trajectories tab, on the preset machine (20 electrons over 40 ns, RF phase lags $-30°$ and $+20°$), against a classical RK4 reference at $5 \times 10^{-6}$ ns over each electron's actual interval in the app. Fitted orders (median energy error, at $-30°$ and $+20°$): Euler 1.00, 1.00; Boris 1.03, 1.33; RK4 0.87, 1.12; RhodoSim RK 1.20, 0.77. Electrons whose fate a hard magnet edge decides are excluded (-30°: 0, +20°: 6). Node v22.22.0, Intel(R) Xeon(R) Processor @ 2.10GHz (2 core), 2026-10-15.

**Findings.**

- At the default step of $10^{-4}$ ns, Boris's energies are typically within 0.005 % of the reference, and RK4's and RhodoSim RK's about as close; Euler's are about 0.1 % high.
- Relative to Boris, a step costs about 1.2 times as much with Euler, 3.5 to 5 times with RhodoSim RK and 5.5 times with RK4.
- On this machine every scheme converges at first order. The half-spheres' fields switch on abruptly at their boundary, and a step that straddles it costs every scheme an error proportional to the step, whatever its formal order (in a uniform field Euler is first order, Boris and RhodoSim RK second, RK4 fourth). The schemes differ by their error constant: Euler's is about a hundred times Boris's, so for Boris's accuracy it needs a step about a hundred times smaller.
- Boris is the sensible default. RK4 gains nothing over it here, for about five times the cost. RhodoSim RK is for results that must agree with C++ RhodoSim (its main branch), which it reproduces step for step; Euler is for comparison and teaching.

# 10 Import/export: configuration and results

**config.ini.** RhodoSim's configuration file, one `key=value` per line; lines starting with `#` are comments, and keys are not case-sensitive. Edit config.ini shows the current settings in this form, applies edited or pasted text, and saves the file. RhodoSim++ reads and writes C++ RhodoSim's keys:

| Key | Unit | Meaning |
|---|---|---|
| `emax` | MV/m | the peak field at the inner conductor (Peak E at R1) |
| `freq` | MHz | the RF frequency |
| `phaselag` | degrees | the RF phase lag |
| `r1`, `r2` | m | the inner and outer conductor radii |
| `magrotation` | degrees | the magnets' entry rotation |
| `addmagnet` | T, m, m | one magnet per line, written `B , R , <dist>`: its field, its radius and its distance from the centre; the lines in a file replace the magnet list |
| `einmean` | MeV | the injection energy (`ein` is read too) |
| `einstd` | MeV | the energy spread $\sigma$ |
| `targeten` | MeV | the top of the energy colour scale (Color scale max) |
| `guntime`, `gunperiod` | ns | the gun's pulse length and pulse period |
| `enum` | | electrons per bunch |
| `bunchnum` | | the number of bunches |
| `starttime`, `endtime` | ns | the start and end of the run |
| `dt` | ns | the time step (it must be positive) |
| `multh`, `thcount` | | C++ RhodoSim's threading: it moves electrons only in its multi-threaded path, so RhodoSim++ writes `multh=1` and `thcount=8` for it, and ignores both when reading |

It adds its own keys, which C++ RhodoSim ignores:

| Key | Unit | Meaning |
|---|---|---|
| `integrator` | | `euler`, `boris`, `rk4` or `rhodosim` |
| `magmodel` | | `sphere` (half-sphere) or `pole` |
| `fringe1G` | m | pole magnets: the distance at which the fringe falls to 1 G |
| `fieldmodel` | | `coax` (analytic) or `solved` (the Cavity tab's field map) |
| `zsig`, `zpsig` | mm, mrad | the gun's vertical size and divergence |
| `portdia` | mm | the port diameter |
| `polegap` | m | the pole gap |
| `beamcurrent` | mA | the gun's average current, for the space-charge guide (0: none) |

A file must contain at least one known key; unknown keys are ignored, so files written by C++ RhodoSim load as they are.

**Links.** Copy link (beside Edit config.ini, and at the bottom of the Cavity tab's side panel beside Copy parameters) copies the page's address with the settings after a `#`, for example

```
rhodosim.html#tab=cavity&cav.H=740&emax=1.1
```

Only the values that differ from the preset are written, so the link opens the same machine for everyone. A link can name the tab (`tab=trajectories`, `analysis`, `sweep` or `cavity`), the view (`mode=top` or `3d`, `zx=1`, `10` or `50`), Trajectories' settings by their config.ini keys (`addmagnet=B,R,dist`, once for each magnet), and the Cavity tab's design as `cav.preset` (`design`, `ideal` or `pillbox`), `cav.R1`, `cav.R2`, `cav.H`, `cav.ro`, `cav.ri`, `cav.t`, `cav.mb`, `cav.h` (mm), `cav.midplane` (`pmc` or `pec`) and `cav.target` (MHz). Opening a link applies its values for that visit only; a bar above the view says so and names any keys it ignored (unknown, or out of range). Keep them makes them your saved settings; Discard reopens the page with your saved settings. A link never starts a run or a solution.

**Field maps, STL and Superfish.** The Cavity tab exports field maps for RhodoSim (text, read by RhodoSim's imported-field model, or binary), in CST's nine-column format, or as the native *r*–*z* grid; the cavity's body as STL; and the cavity as Superfish input (an Automesh file). Reading Superfish's own field maps back is not supported yet.

# 11 Checking the results

Both halves of RhodoSim++ carry a self-test that you can run in your own browser. Each check compares the program with an *independent* oracle (an exact solution, a conservation law, a separate calculation, or recorded results) and shows the measured and expected values, the error and the tolerance.

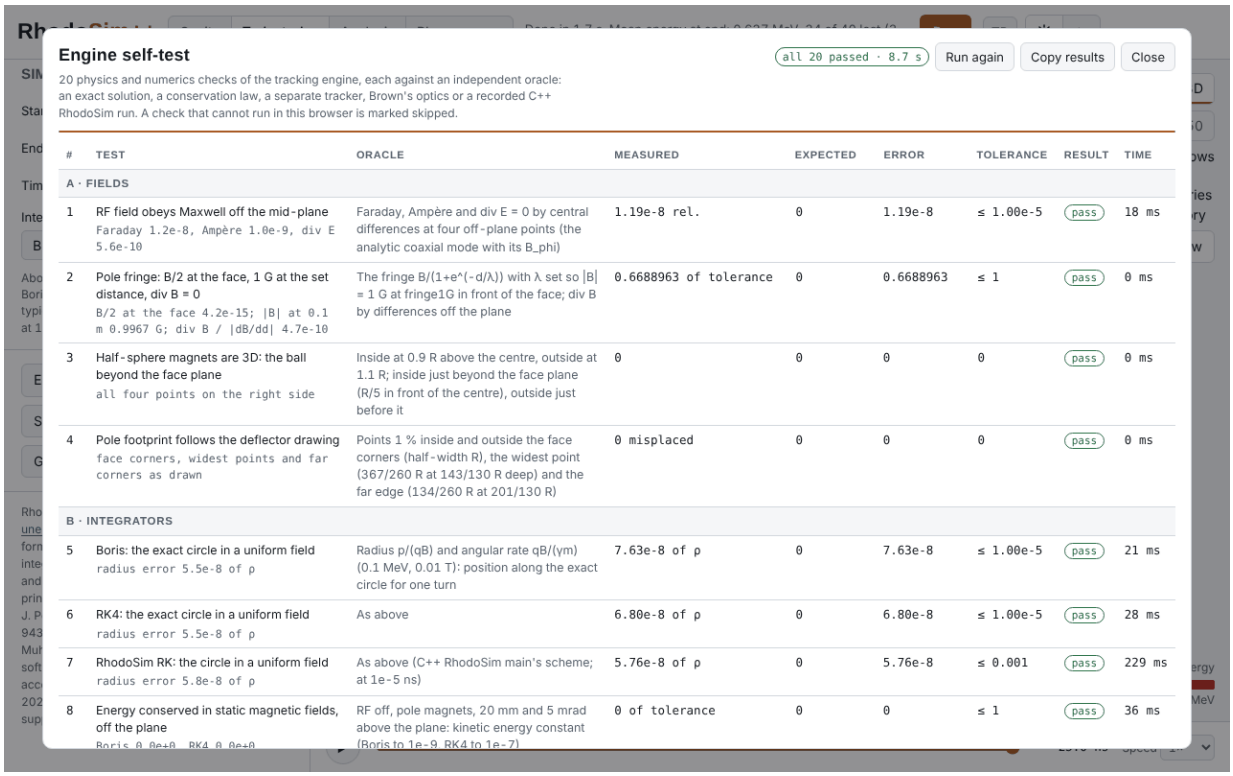



**Figure 15:** The engine's self-test (Self-test in the Simulation section).

- **The engine's self-test** (Self-test in the Simulation section): 20 checks in six groups: fields (Maxwell's equations, the magnet shapes), integrators (exact motion in a uniform field, energy conservation, orders of convergence), acceleration (an independent tracker), 3D motion (symmetry, edge focusing), losses (exact times), and consistency (Web Workers against the main thread, `config.ini`, and C++ RhodoSim's recorded results).
- **The cavity solver's self-test** (Self-test in the Cavity tab): 49 checks of the geometry, the mesh, the solver against exact modes, perturbation theory, the field identities, the export and mesh convergence.

A check that cannot run in your browser is marked *skipped* rather than passed.